PV102 Day 8: Commissioning Troubleshooting and Professional Handoff

Students close PV102 by verifying a system from evidence: pre-energization inspection, polarity and continuity checks, voltage checks, functional checks, fault response, documentation, and client handoff.

PV102 | CTS Day 24 | Instructor Run Sheet
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Core rule

Test, document, and hand off what the system actually does.

Learning targets

- Use a commissioning sequence to verify before energizing.
- Interpret basic test results without guessing beyond evidence.
- Troubleshoot from symptom to boundary to measurement.
- Prepare a concise handoff that names what passed, what failed, what changed, and what remains unknown.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Training kits or paper fault cards Multimeters Commissioning checklist Handoff form Labels Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 7 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.

12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.
1:00-2:15	Commissioning and Fault Handoff Challenge	Teams inspect a training system or fault card set, run a commissioning sequence, identify one fault, and deliver a final technical handoff.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 is complete. Carry the habit forward: define the job, verify the path, protect the boundary, document the truth.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use commissioning troubleshooting and professional handoff to make the next design decision safer and more defensible. The rule is simple: Test, document, and hand off what the system actually does."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Review the system scope and safety boundary.
- Complete pre-energization inspection using the checklist.
- Run only authorized de-energized checks first.
- Use evidence to identify the fault or missing documentation.
- Record pass/fail results and corrective action recommendation.
- Deliver a three-minute final handoff: system job, path, checks, fault, fix/next action, unknowns.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 is complete. Carry the habit forward: define the job, verify the path, protect the boundary, document the truth.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 8 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 24 | Morning Quiz - Answer Key
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. Commissioning means:

- A. Verifying a system before/while placing it into service
- B. Buying the biggest inverter
- C. Painting a roof
- D. Guessing until it works

Answer: A

Rationale/look-for: Commissioning uses inspection and tests to prove readiness.

2. Before continuity testing, the circuit must be:

- A. De-energized and verified
- B. Fully energized
- C. Connected to utility
- D. Under maximum load

Answer: A

Rationale/look-for: Continuity/resistance checks require a safe de-energized condition.

3. Polarity checks help prevent:

- A. Reverse connection damage or unsafe operation
- B. Too much sunlight
- C. Roof leaks only
- D. Bad handwriting

Answer: A

Rationale/look-for: Correct polarity is a basic commissioning requirement.

4. A professional handoff should include:

- A. Results, changes, unknowns, and next actions
- B. Only “it works”
- C. No measurements
- D. Guesswork

Answer: A

Rationale/look-for: Handoff documentation makes the next person safer and more effective.

5. Good troubleshooting starts by:

- A. Defining the symptom and boundary
- B. Replacing random parts
- C. Blaming the battery immediately
- D. Skipping measurements

Answer: A

Rationale/look-for: Symptom and boundary focus measurements.

6. A 12 V battery reads 0 V at the load terminals. Name one possible boundary to check.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept fuse, breaker, switch, connector, polarity, wire, controller output, load terminal.

7. Why should failed tests be documented instead of hidden?

_____

_____

Answer: Safety/learning

Rationale/look-for: Failed tests guide next action and prevent unsafe assumptions.

8. What should happen if a measurement does not make sense?

_____

_____

Answer: Stop/verify

Rationale/look-for: Stop, verify meter setup, verify reference points, ask for review, do not guess.

9. Name one item that belongs in a final PV102 handoff package.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept diagram, load table, site notes, commissioning checklist, measurements, photos, unknowns, next steps.

10. Write the Day 8 rule in your own words.

_____

_____

Answer: Test and handoff

Rationale/look-for: Look for verification and documentation.

PV102 Day 8 Whiteboard Outline

PV102 | CTS Day 24 | Whiteboard Outline

Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Commissioning Troubleshooting and Professional Handoff Rule: Test, document, and hand off what the system actually does. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- Training kit has a hidden open fuse and incomplete label.
- Paper system has reversed polarity at one component card and missing OCPD label.
- Mock client handoff has incomplete load table and no record of final voltage checks.

Final board sentence

Test, document, and hand off what the system actually does.

PV102 Day 8 Student Packet

Commissioning Troubleshooting and Professional Handoff

PV102 | CTS Day 24 | Student Print Packet
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Today's rule

Test, document, and hand off what the system actually does.

Learning targets

- Use a commissioning sequence to verify before energizing.
- Interpret basic test results without guessing beyond evidence.
- Troubleshoot from symptom to boundary to measurement.
- Prepare a concise handoff that names what passed, what failed, what changed, and what remains unknown.

Morning quiz

1. Commissioning means:

- A. Verifying a system before/while placing it into service
- B. Buying the biggest inverter
- C. Painting a roof
- D. Guessing until it works

2. Before continuity testing, the circuit must be:

- A. De-energized and verified
- B. Fully energized
- C. Connected to utility
- D. Under maximum load

3. Polarity checks help prevent:

- A. Reverse connection damage or unsafe operation
- B. Too much sunlight
- C. Roof leaks only
- D. Bad handwriting

4. A professional handoff should include:

- A. Results, changes, unknowns, and next actions
- B. Only “it works”
- C. No measurements
- D. Guesswork

5. Good troubleshooting starts by:

- A. Defining the symptom and boundary
- B. Replacing random parts
- C. Blaming the battery immediately
- D. Skipping measurements

6. A 12 V battery reads 0 V at the load terminals. Name one possible boundary to check.

_____

_____

7. Why should failed tests be documented instead of hidden?

_____

_____

8. What should happen if a measurement does not make sense?

_____

_____

9. Name one item that belongs in a final PV102 handoff package.

_____

_____

10. Write the Day 8 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Commissioning and Fault Handoff Challenge

Teams inspect a training system or fault card set, run a commissioning sequence, identify one fault, and deliver a final technical handoff.

Scenario options

- Training kit has a hidden open fuse and incomplete label.
- Paper system has reversed polarity at one component card and missing OCPD label.
- Mock client handoff has incomplete load table and no record of final voltage checks.

Team procedure

- Review the system scope and safety boundary.
- Complete pre-energization inspection using the checklist.
- Run only authorized de-energized checks first.
- Use evidence to identify the fault or missing documentation.
- Record pass/fail results and corrective action recommendation.
- Deliver a three-minute final handoff: system job, path, checks, fault, fix/next action, unknowns.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

What is one thing you now refuse to accept as “good enough” without evidence?

_____

_____

PV102 Day 8 Afternoon Lab

Commissioning and Fault Handoff Challenge

PV102 | CTS Day 24 | Afternoon Lab
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Mission

Teams inspect a training system or fault card set, run a commissioning sequence, identify one fault, and deliver a final technical handoff.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Training kit has a hidden open fuse and incomplete label.
- Paper system has reversed polarity at one component card and missing OCPD label.
- Mock client handoff has incomplete load table and no record of final voltage checks.

Procedure

- Review the system scope and safety boundary.
- Complete pre-energization inspection using the checklist.
- Run only authorized de-energized checks first.
- Use evidence to identify the fault or missing documentation.
- Record pass/fail results and corrective action recommendation.
- Deliver a three-minute final handoff: system job, path, checks, fault, fix/next action, unknowns.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	

Primary decision	
Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 8 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 24 | Google Classroom Post
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Announcement

Today in PV102 we are working on commissioning troubleshooting and professional handoff. The core rule is: Test, document, and hand off what the system actually does.

Assignment title

PV102 Day 8 - Commissioning Troubleshooting and Professional Handoff

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.